

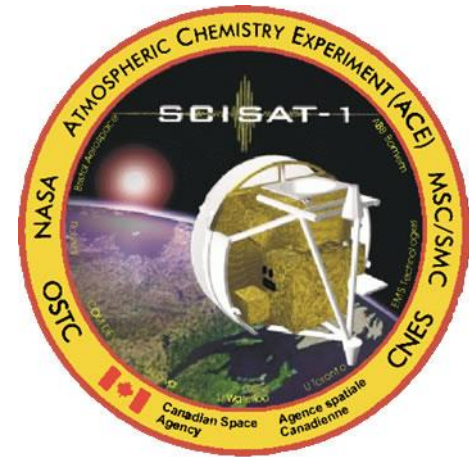
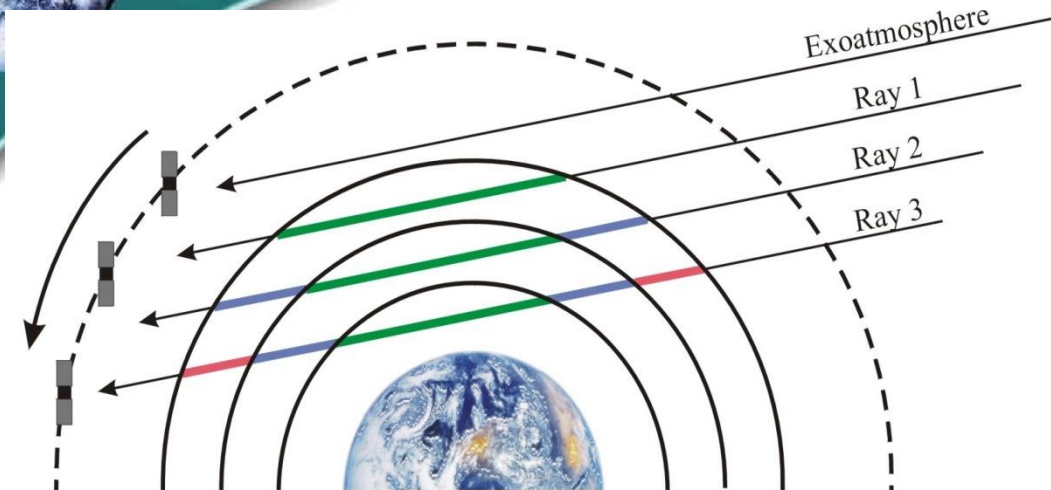
New and Improved Infrared Spectroscopy of Halogen-Containing Species for ACE-FTS Retrievals

Jeremy J. Harrison

University of York



Atmospheric Chemistry Experiment (ACE)



- FTS: 750 to 4400 cm^{-1}
- Radiance of sun gives high S/N
- Long pathlengths ~ 300 km (limb sounding)
- Measurements at many altitudes
- ACE detects more trace organic molecules than any other satellite instrument.

Halocarbons

- Strong greenhouse gases
- Many are ozone depleting substances
 - Regulated by the Montreal Protocol
- Many are increasing in the atmosphere
- Many have long atmospheric lifetimes

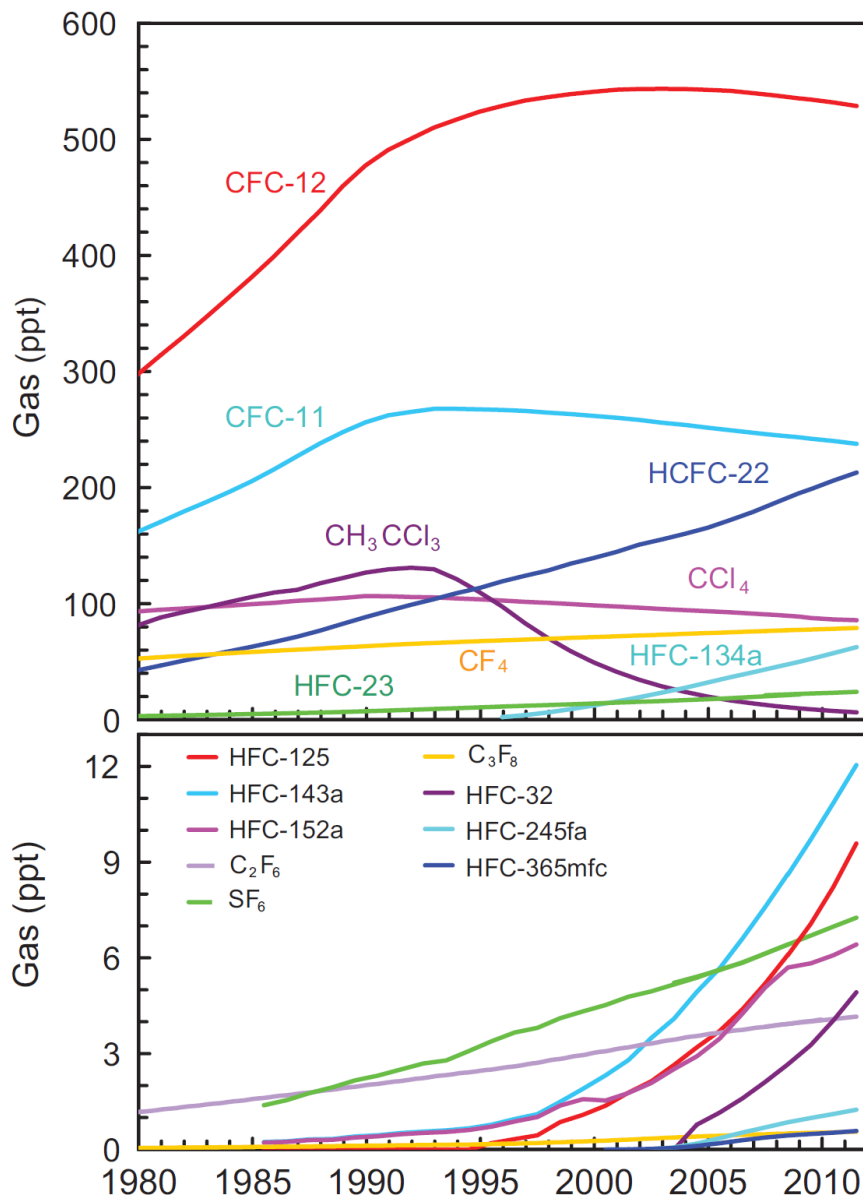


Figure 2.4 | Globally averaged dry-air mole fractions at the Earth's surface of the major halogen-containing well-mixed GHG. These are derived mainly using monthly mean measurements from the AGAGE and NOAA/ESRL/GMD networks. For clarity, only the most abundant chemicals are shown in different compound classes and results from different networks have been combined when both are available.



Laboratory Spectroscopy

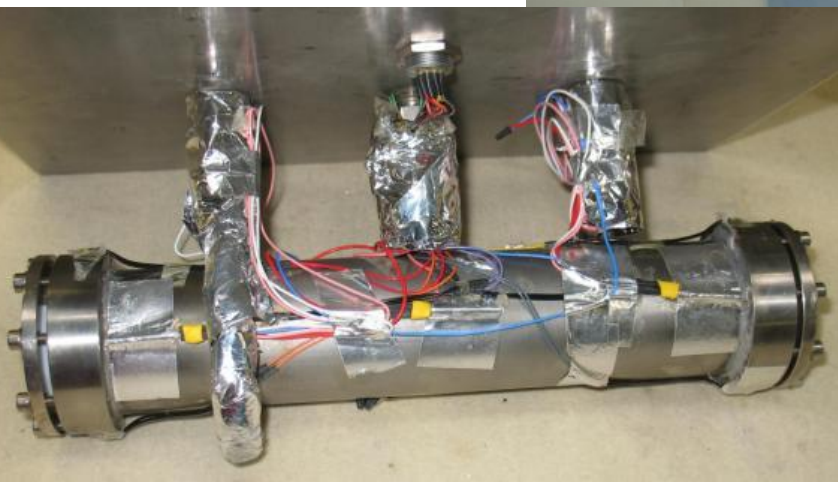
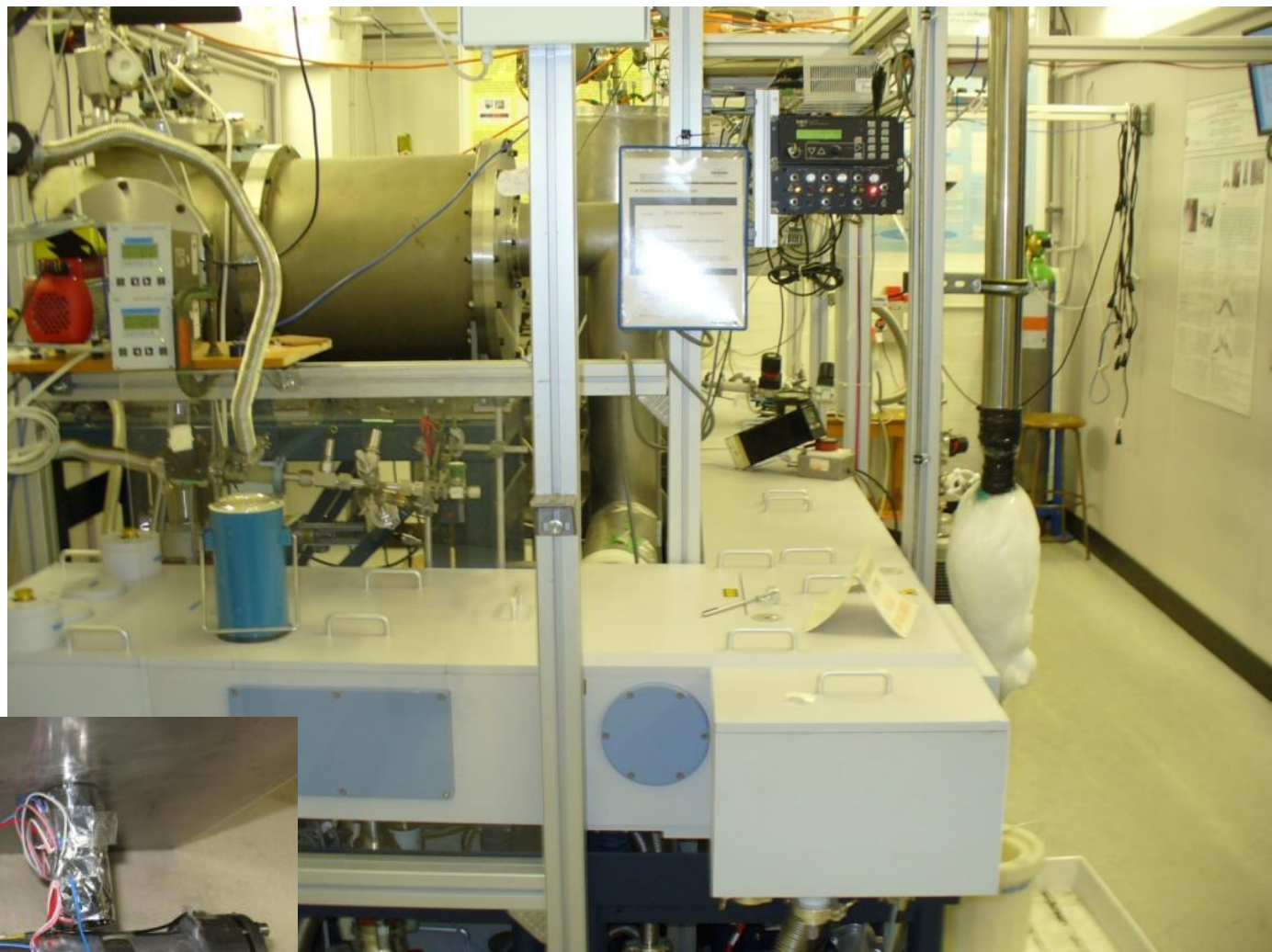
- HITRAN contains a number of absorption cross section datasets for Cl- and F-containing molecules
- Dataset requirements for remote sensing:
 - Accurate band intensities
 - Accurate wavenumber scale
 - Wide P-T coverage for best interpolation
 - Appropriate resolution (Doppler-limited)
- A number of HITRAN datasets fail in many of these respects.



MSF at RAL

Bruker IFS
125 HR

26-cm
absorption cell





Trifluoromethane (HFC-23)

- Hydrofluorocarbons (HFCs) are replacements for CFCs and HCFCs
 - Not regulated by Montreal Protocol
- HFC-23 formed as a by-product of HCFC-22 (CHClF_2) production
- Second most abundant HFC in atmosphere
- Lifetime of 222 years
- 100-year GWP of 14,200
- Globally averaged HFC-23 mixing ratios reached 21.8 ppt in 2008, with a yearly increase of 0.8 ppt/yr (3.9%/yr).



Previous HFC-23 absorption cross sections



Journal of Quantitative Spectroscopy &
Radiative Transfer 96 (2005) 281–287

Journal of
Quantitative
Spectroscopy &
Radiative
Transfer

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Absorption cross-sections of HFC-23 at atmospheric conditions

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Received 30 December 2003; received in revised form 15 April 2005; accepted 18 April 2005

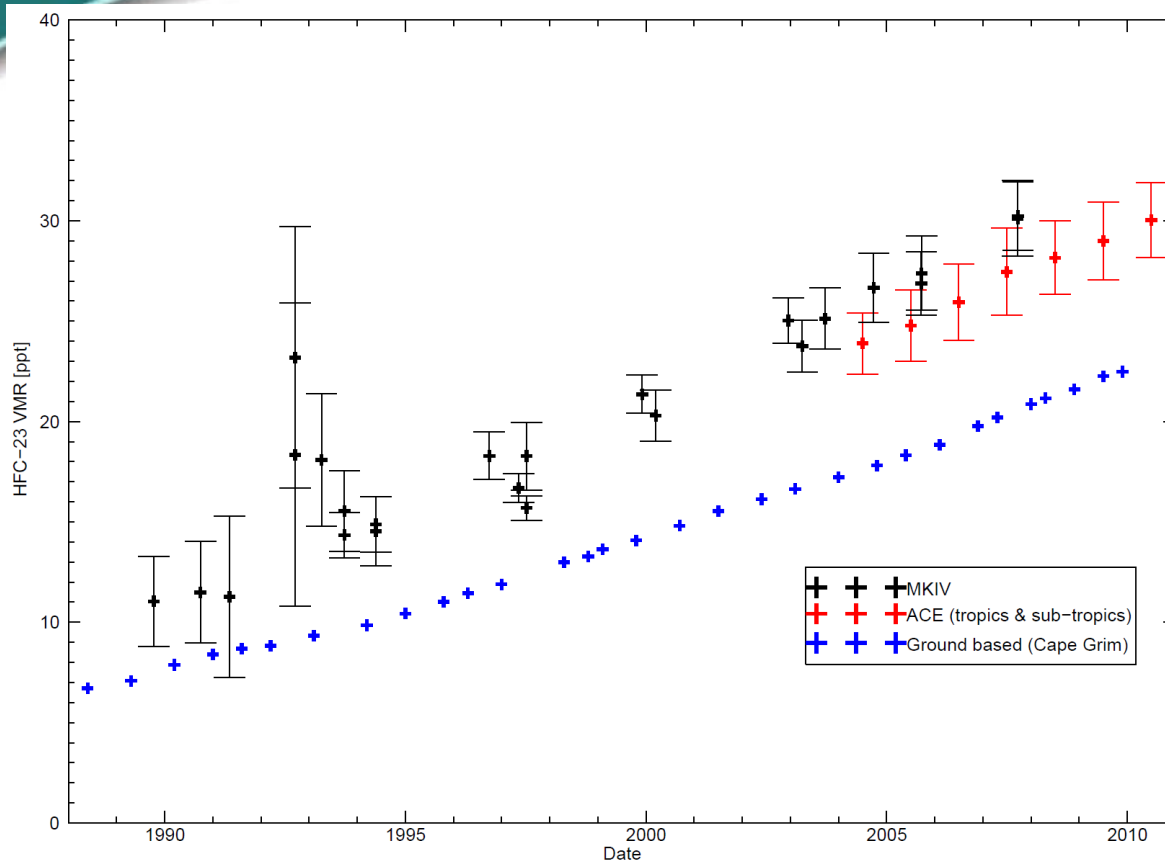
Abstract

Spectral absorption cross-sections, k_ν ($\text{cm}^{-1} \text{atm}^{-1}$), were measured in the $8.64 \mu\text{m}$ band system of HFC-23 using a high-resolution Fourier transform spectrometer. Temperature and total (N_2 -broadening) pressure were selected to represent atmospheric layers according to various atmospheric models. The measured absolute intensity (or integrated cross-section) was obtained to be $0.693 \pm 0.029 \times 10^{-16} \text{cm molecule}^{-1}$.

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Keywords: Absorption cross-sections; HFC-23; Thermal infrared

First remote-sensing observations of HFC-23

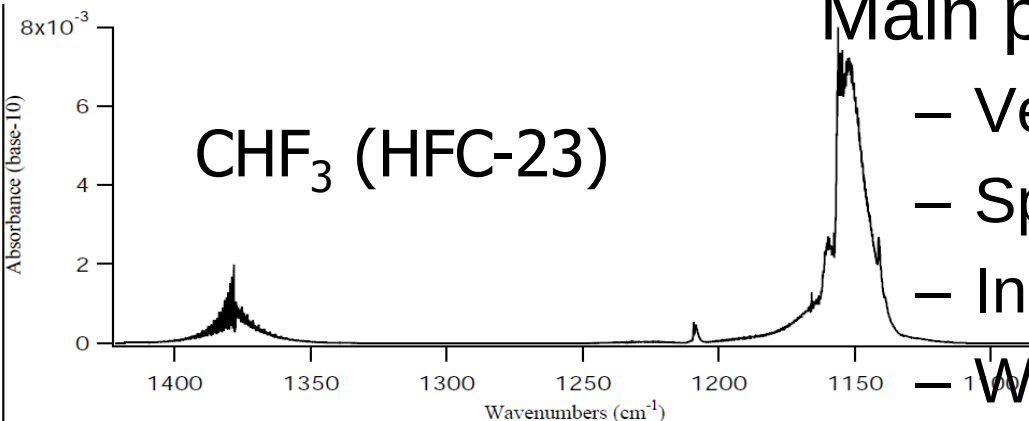
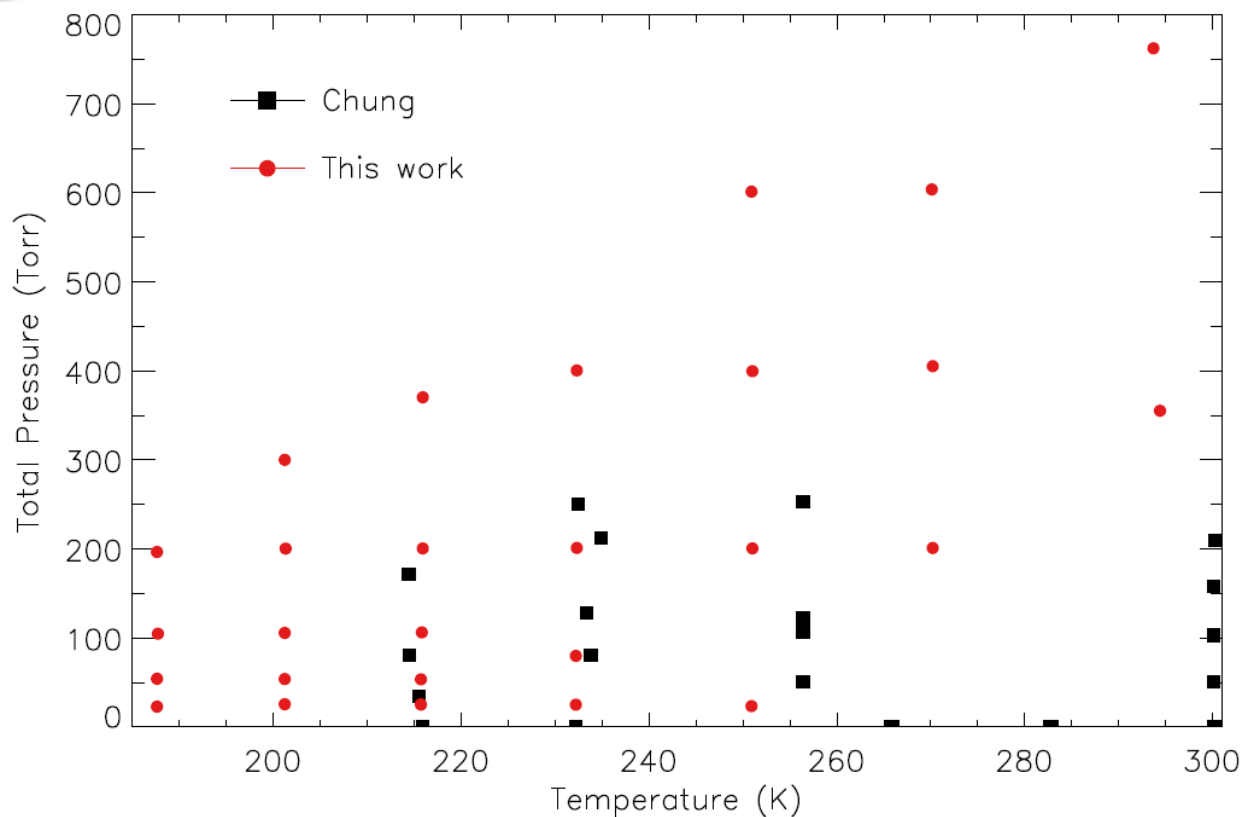


Harrison et al., First remote-sensing observations of trifluoromethane (HFC-23) in the upper troposphere and lower stratosphere, J. Geophys. Res., 2012, 117, D05308, doi:10.1029/2011JD016423.



CHF₃ cross sections

New dataset:
Harrison, JQSRT
130, 359-364
(2013)



Main problems with old dataset:

- Very poor P-T coverage
- Spectra under-resolved
- Inconsistent baselines

– Wavenumber calibration error



1,1,1,2-Tetrafluoroethane (HFC-134a)

- The preferred refrigerant in refrigeration and mobile air conditioning (replacement for CFC-12)
- Most abundant HFC in atmosphere
- Lifetime of 13.4 years
- 100-year GWP of 1,370
- HFC-134a has been growing steadily and reached a global mean mole fraction of ~48 ppt in 2008, with an average trend of 4.6 ppt/yr (~10%/yr) in 2007–2008.



Remote-sensing observations of HFC-134a



JOURNAL OF GEOPHYSICAL RESEARCH, VOL. 111, D22313, doi:10.1029/2006JD007395, 2006

A global inventory of stratospheric fluorine in 2004 based on Atmospheric Chemistry Experiment Fourier transform spectrometer (ACE-FTS) measurements

R. Nassar,^{1,2} P. F. Bernath,¹ C. D. Boone,¹ S. D. McLeod,¹ R. Skelton,¹ K. A. Walker,^{1,3} C. P. Rinsland,⁴ and P. Duchatelet⁵

Received 11 April 2006; revised 29 June 2006; accepted 7 August 2006; published 30 November 2006.

[1] Total fluorine (F_{TOT}) in the stratosphere has been determined using Atmospheric Chemistry Experiment Fourier transform spectrometer (ACE-FTS) measurements of HF, COF₂, COClF, CF₄, CCl₃F (CFC-11), CCl₂F₂ (CFC-12), CHClF₂ (HCFC-22), CCl₂FCClF₂ (CFC-113), CH₃CClF₂ (HCFC-142b), CH₂FCF₃ (HFC-134a), and SF₆. The retrieval of HFC-134a (CH₂FCF₃) from spaceborne measurements had not been carried out prior to this work. Measurements of these species have been supplemented by data from models to extend the altitude range of the profiles and have also been complemented by estimates of 15 minor fluorine species. Using these data, separate fluorine budgets were determined in five latitude zones (60°–82°N, 30°–60°N, 30°S–30°N, 30°–60°S, and 60°–82°S) by averaging over the period of February 2004 to January 2005 inclusive, when possible. Stratospheric F_{TOT} profiles in each latitude zone are nearly linear, with mean stratospheric F_{TOT} values ranging from 2.50 to 2.59 ppbv (with a 1σ precision of 0.04–0.07 ppbv and an estimated accuracy of 0.15 ppbv) for each zone. The highest mean F_{TOT} value occurred in the tropics, which is qualitatively consistent with increasing levels of stratospheric fluorine and the mean stratospheric circulation pattern.

Citation: Nassar, R., P. F. Bernath, C. D. Boone, S. D. McLeod, R. Skelton, K. A. Walker, C. P. Rinsland, and P. Duchatelet (2006), A global inventory of stratospheric fluorine in 2004 based on Atmospheric Chemistry Experiment Fourier transform spectrometer (ACE-FTS) measurements, *J. Geophys. Res.*, 111, D22313, doi:10.1029/2006JD007395.

Nassar et al., *J. Geophys. Res.*, 2006, 111, D22313,
doi:10.1029/2006JD007395.

HFC-134a absorption cross sections in HITRAN



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Journal of Quantitative Spectroscopy &
Radiative Transfer 83 (2004) 285–294

Journal of
Quantitative
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Radiative
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Absorption cross-sections of HFC-134a in the spectral region between 7 and 12 μm

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Received 13 February 2002; received in revised form 11 November 2002; accepted 13 November 2002

Abstract

Spectral absorption cross-sections k_ν ($\text{cm}^2 \text{ molecule}^{-1}$) were measured in the 7.07, 8.08, 9.24, 10.3, and 11.9 μm bands of HFC-134a at temperatures between 190 and 296 K and pressures between 10 and 1013 hPa with a high-resolution Fourier-transform spectrometer. Our data were obtained at pressures and temperatures that characterise several of the commonly adapted model atmospheres, represent tangent heights in solar-occultation-type remote-sensing observations of the atmosphere, and the conditions recorded in atmospheric observations at Arctic latitudes. The integrated cross-sections (or integrated intensities) of the absorption bands in the 7.07, 8.08, 9.24, 10.3, and 11.9 μm regions are, in the unit of $10^{-17} \text{ cm molecule}^{-1}$, 0.4674, 9.711, 1.508, 0.8753, and 0.2527, respectively.

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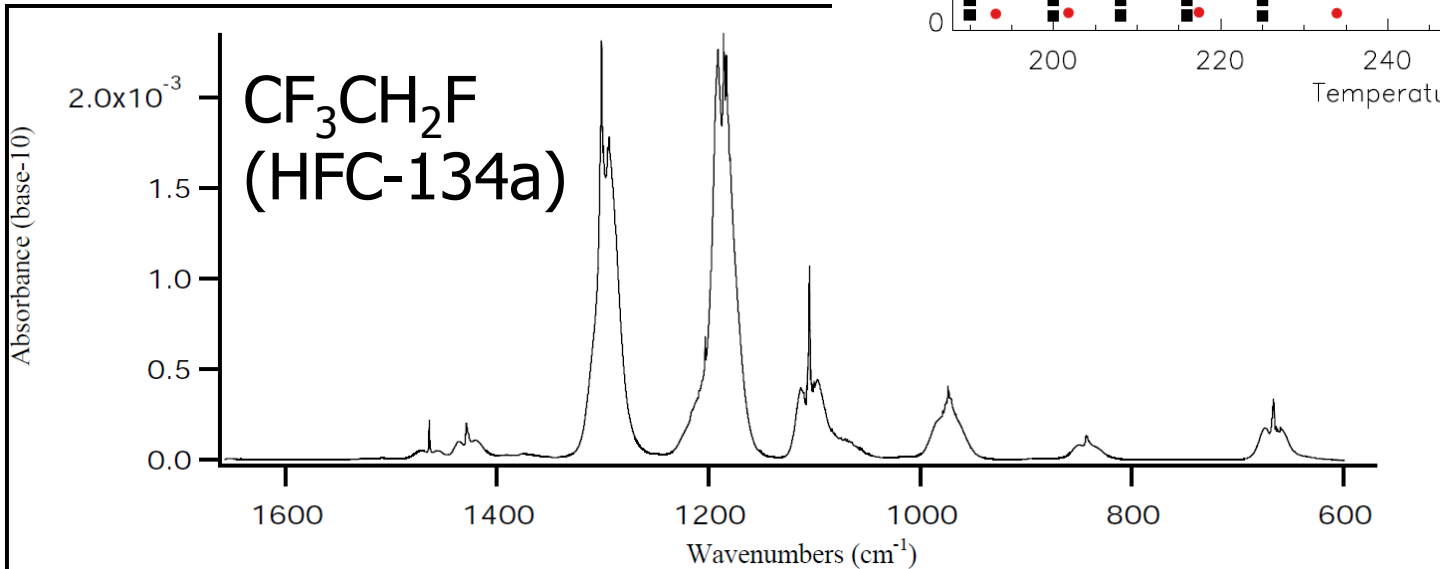
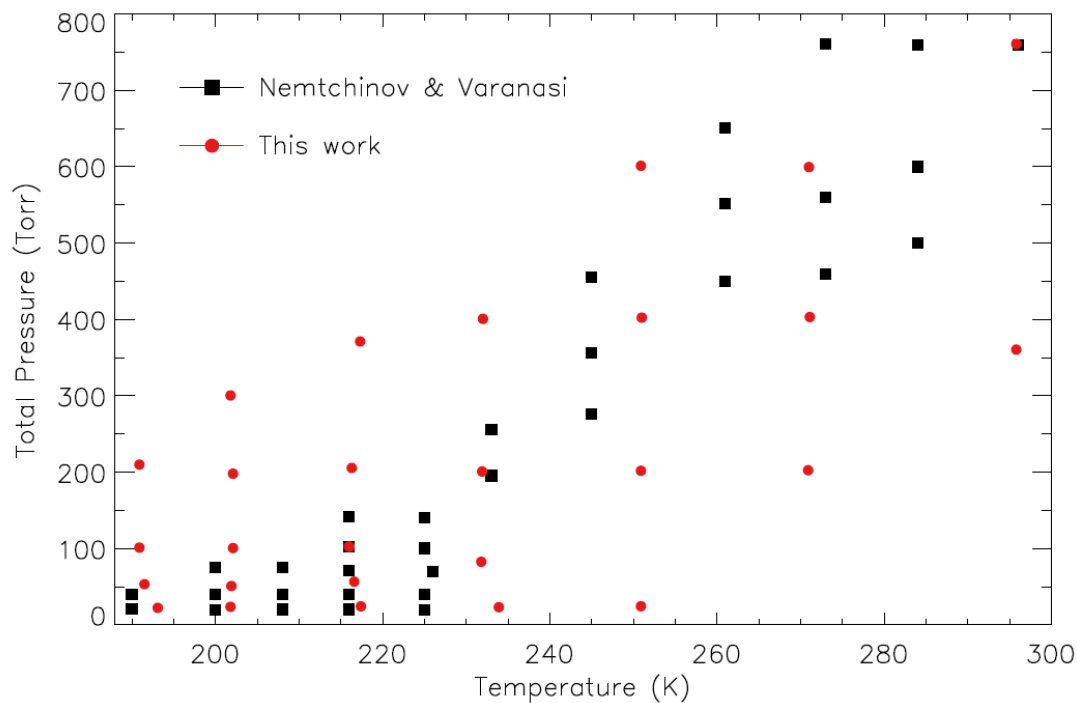
Keywords: HFC-134a; $\text{CH}_2\text{F}-\text{CF}_3$; Thermal IR cross-sections



$\text{CF}_3\text{CH}_2\text{F}$ cross sections

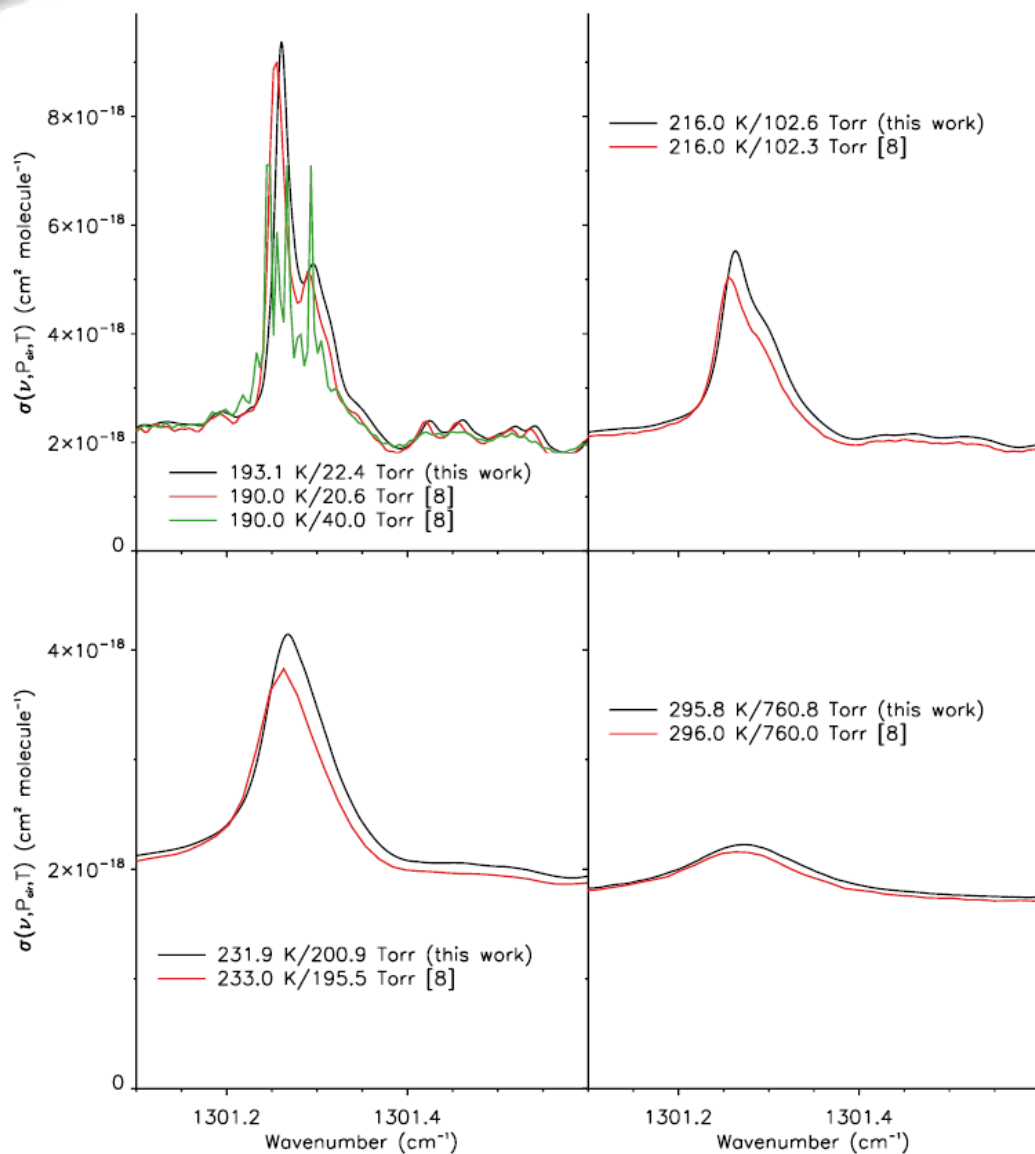
Main problems with old dataset:

- Poor P-T coverage
- Wavenumber calibration error





$\text{CF}_3\text{CH}_2\text{F}$ cross sections



CFC-12 (CCl_2F_2)

- Most abundant anthropogenic halocarbon in the atmosphere
- Was used in refrigeration and air conditioning
- Banned under the Montreal Protocol
- Atmospheric lifetime ~ 100 years

MIPAS:

Kellmann et al., ACP, 12, 11857–11875, 2012

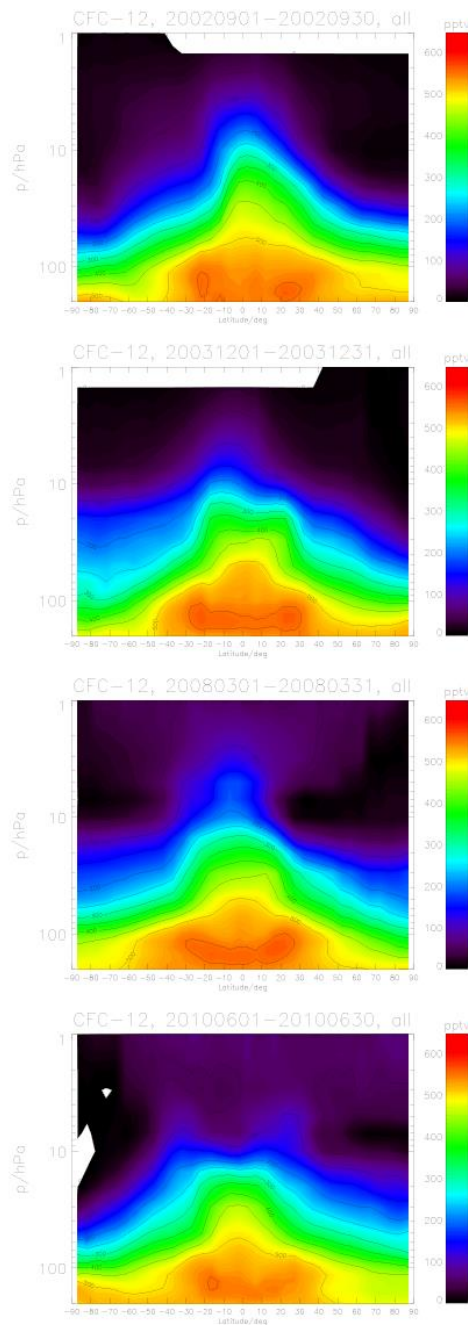


Fig. 8. CFC-12 monthly zonal mean distribution between 200 hPa and 1 hPa in September 2002, December 2003, March 2008, and June 2010.



CFC-12 absorption cross sections in HITRAN



Pergamon

J. Quant. Spectrosc. Radiat. Transfer Vol. 51, No. 5, pp. 679–687, 1994

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0022-4073(94)E0001-M

THERMAL INFRARED ABSORPTION COEFFICIENTS OF CFC-12 AT ATMOSPHERIC CONDITIONS

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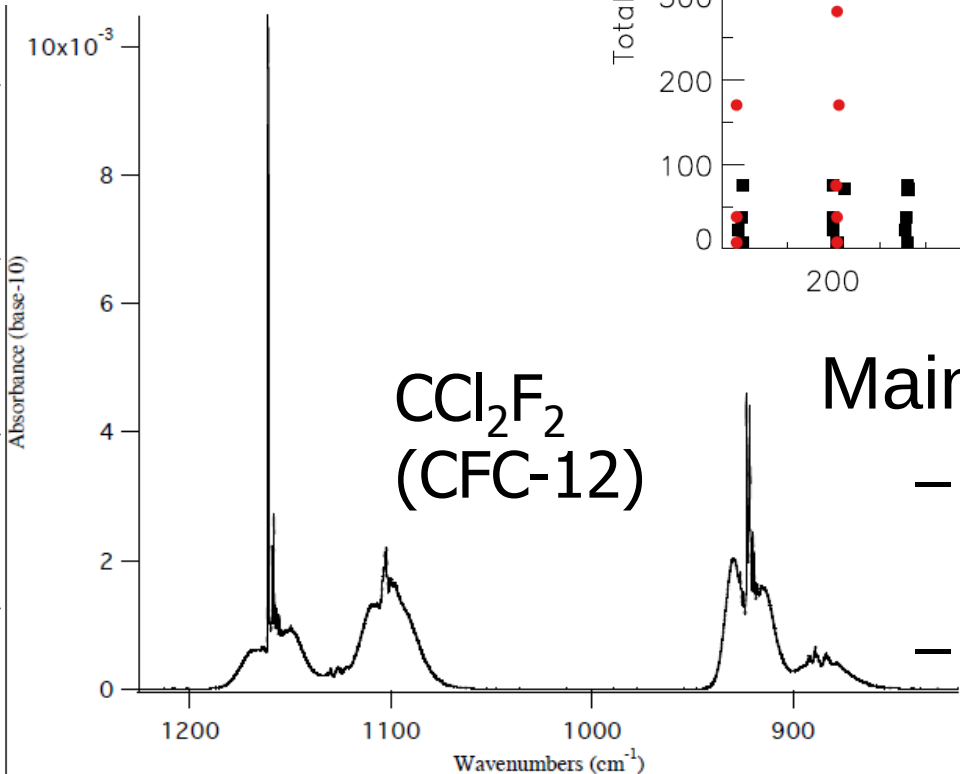
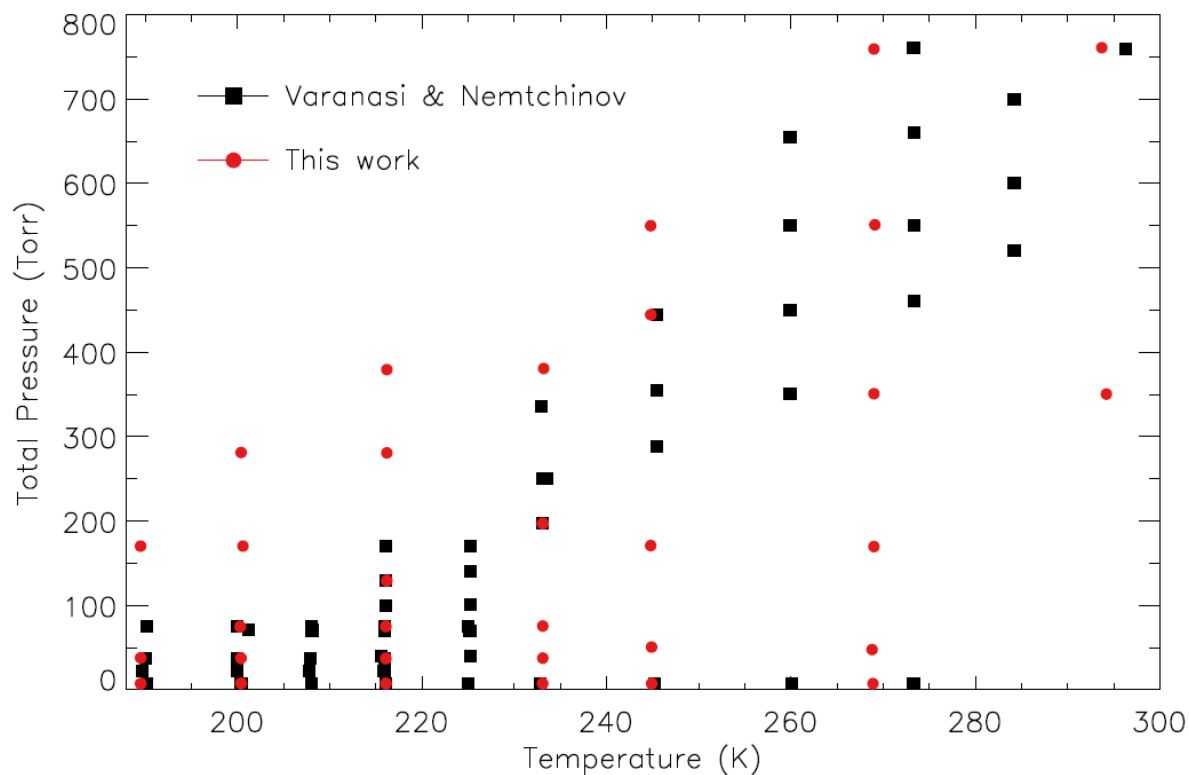
Institute for Terrestrial and Planetary Atmospheres, Marine Sciences Research Center, State University of New York at Stony Brook, Stony Brook, NY 11794-5000, U.S.A.

(Received 4 October 1993)

Abstract—Spectral absorption coefficients k_ν ($\text{cm}^{-1} \text{atm}^{-1}$) have been measured in the 9 and 11 μm bands of CFC-12 (CF_2Cl_2) using a high-resolution Fourier transform spectrometer. Temperature and total (broadening) pressure have been varied to obtain results at conditions representative of tropospheric and stratospheric layers of the atmosphere. The measured absolute intensities (in units of $10^{-17} \text{cm.molecule}^{-1}$) of the 9 and 11 μm bands are 7.595 ± 0.070 and 5.750 ± 0.068 , respectively.



CCl_2F_2 cross sections



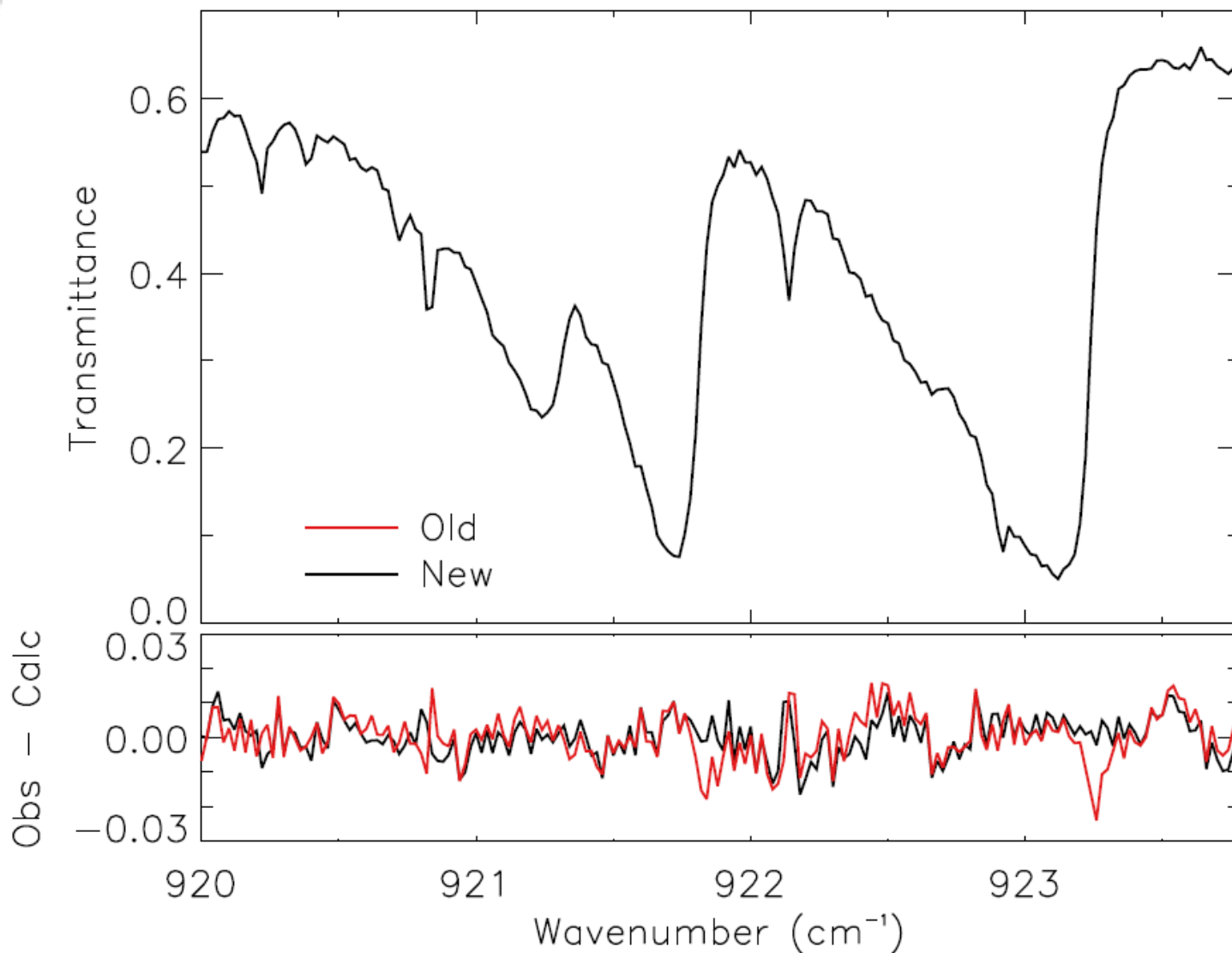
Main problems with old dataset:

- Strong Q branch near 1160 cm^{-1} often saturated.
- Wavenumber calibration error



CCl_2F_2 cross sections

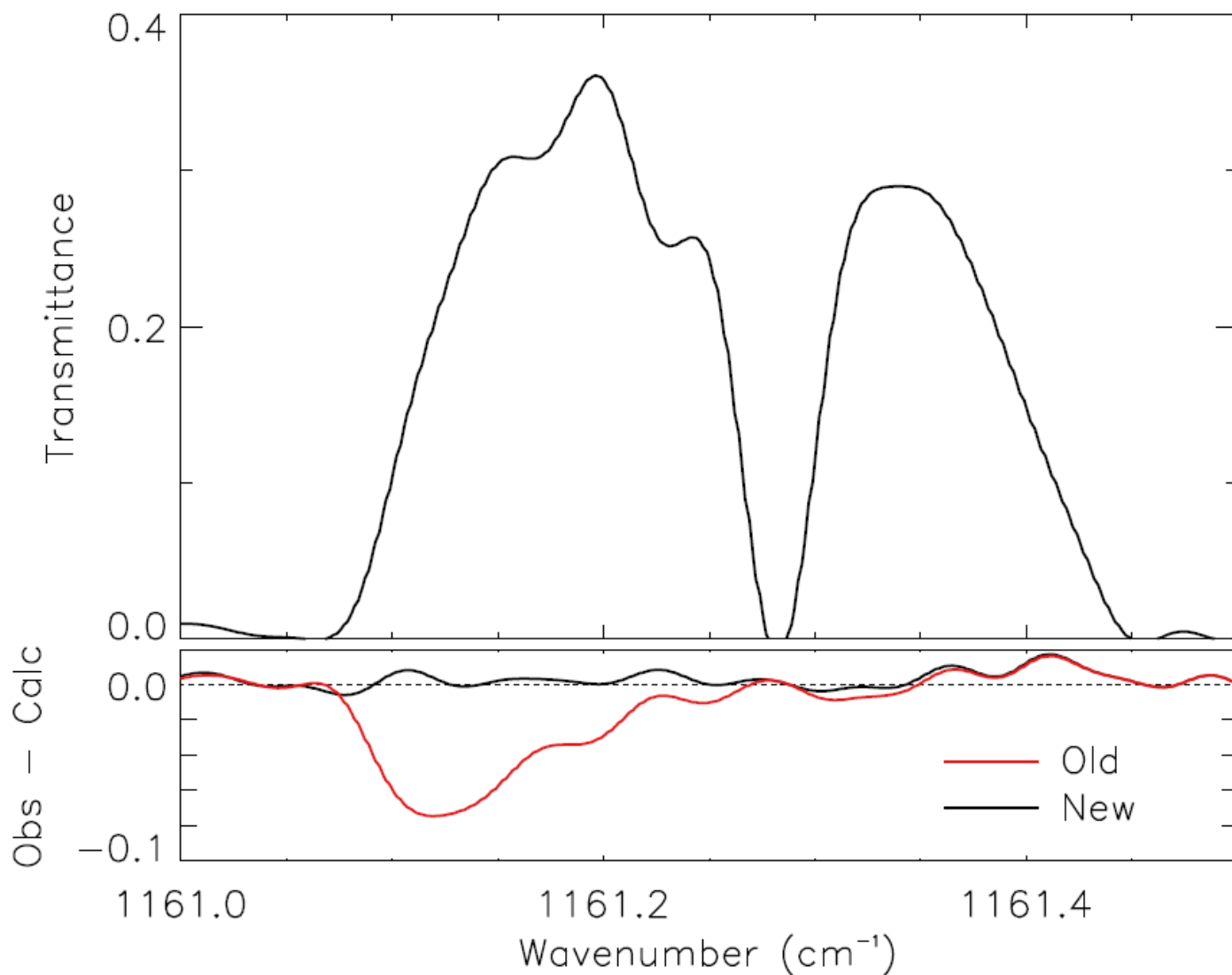
ACE Occultation ss11950: 7.4 km





CCl_2F_2 cross sections

ACE Occultation ss11950: 7.4 km





Carbon tetrachloride

- CCl_4 accounted for 359 ppt (about 11%) of total tropospheric Cl in 2008
- CCl_4 tropospheric abundances have declined less rapidly than expected
 - Top-down emissions 40–80 Gg/yr (2005–2008)
 - Bottom-up emissions 0–30 Gg/yr (2005–2008)
- Errors in reporting, errors in the analysis of reported data, and/or unknown sources?
- Atmospheric lifetime 26 years
- ACE-FTS CCl_4 data product biased ~20% high



CCl_4 absorption cross sections in HITRAN



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Journal of Quantitative Spectroscopy &
Radiative Transfer 82 (2003) 473–481

Journal of
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www.elsevier.com/locate/jqsrt

Thermal infrared absorption cross-sections of CCl_4 needed for atmospheric remote sensing

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NY 11794-5000, USA*

Received 28 July 1998; received in revised form 10 November 2002; accepted 14 November 2002

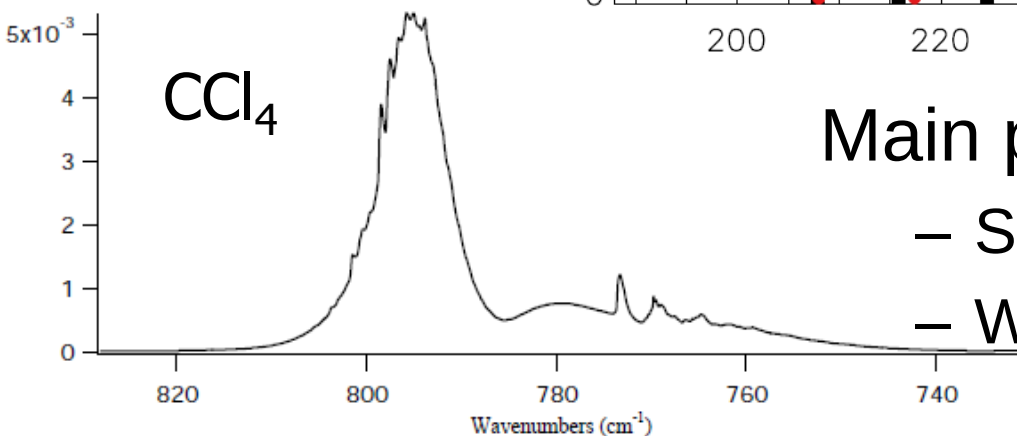
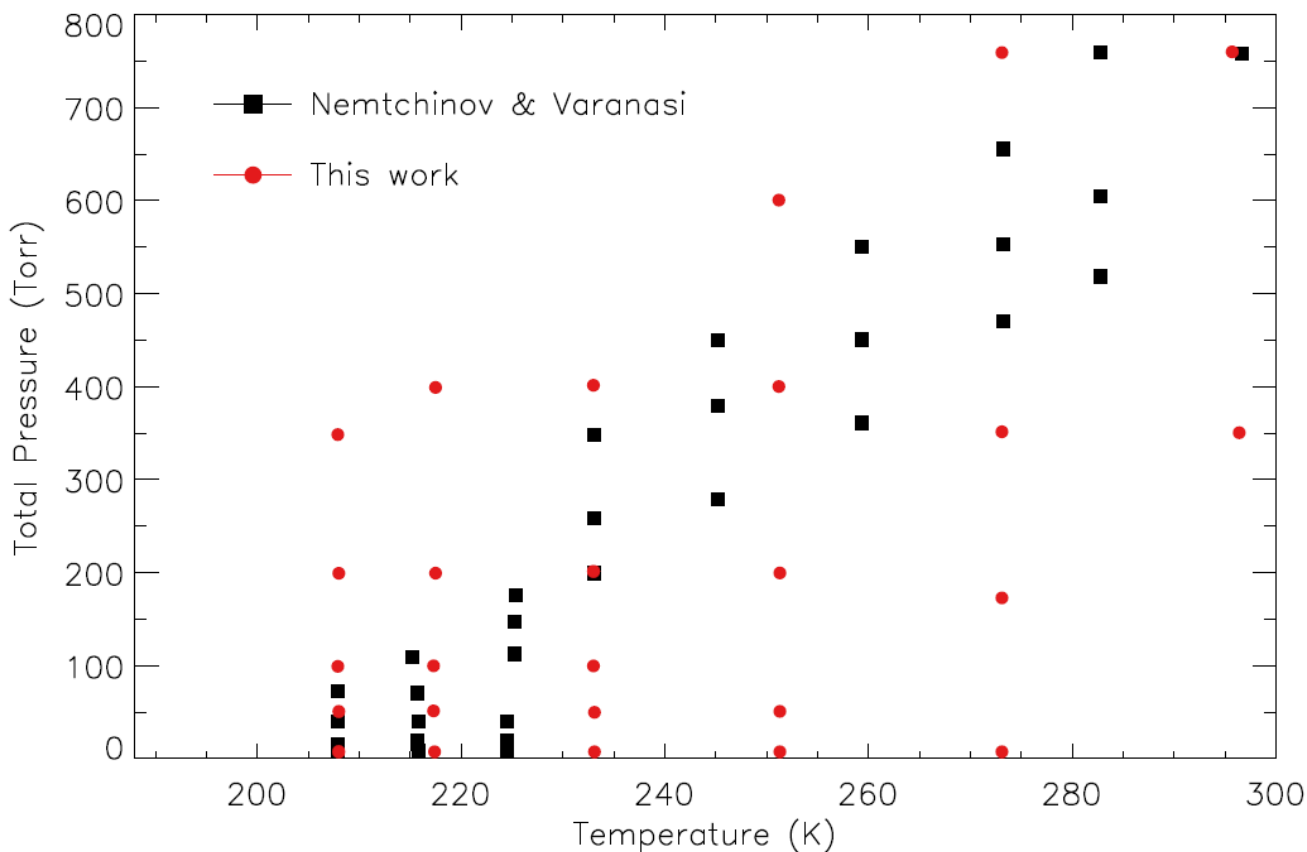
Abstract

Highly accurate spectral absorption cross-sections k_ν ($\text{cm}^2 \text{ molecule}^{-1}$) were measured in the $12.7 \mu\text{m}$ band of CCl_4 at temperatures between 208 and 296 K and pressures between 10 and 1013 mb with a high-resolution Fourier-transform spectrometer. Our data were obtained at pressures p and temperatures T that characterize the atmospheric layers in the commonly adapted models of the atmosphere. The selected (p, T) combinations also represent tangent heights in solar-occultation-type remote sensing observations and actual conditions recorded by balloon-borne instruments in the Arctic. The combined absolute intensity (or integrated cross-section) of the bands contributing to the absorption around $12.7 \mu\text{m}$ is $6.295 \pm 0.102 \times 10^{-17} \text{ cm molecule}^{-1}$.

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CCl_4 cross sections

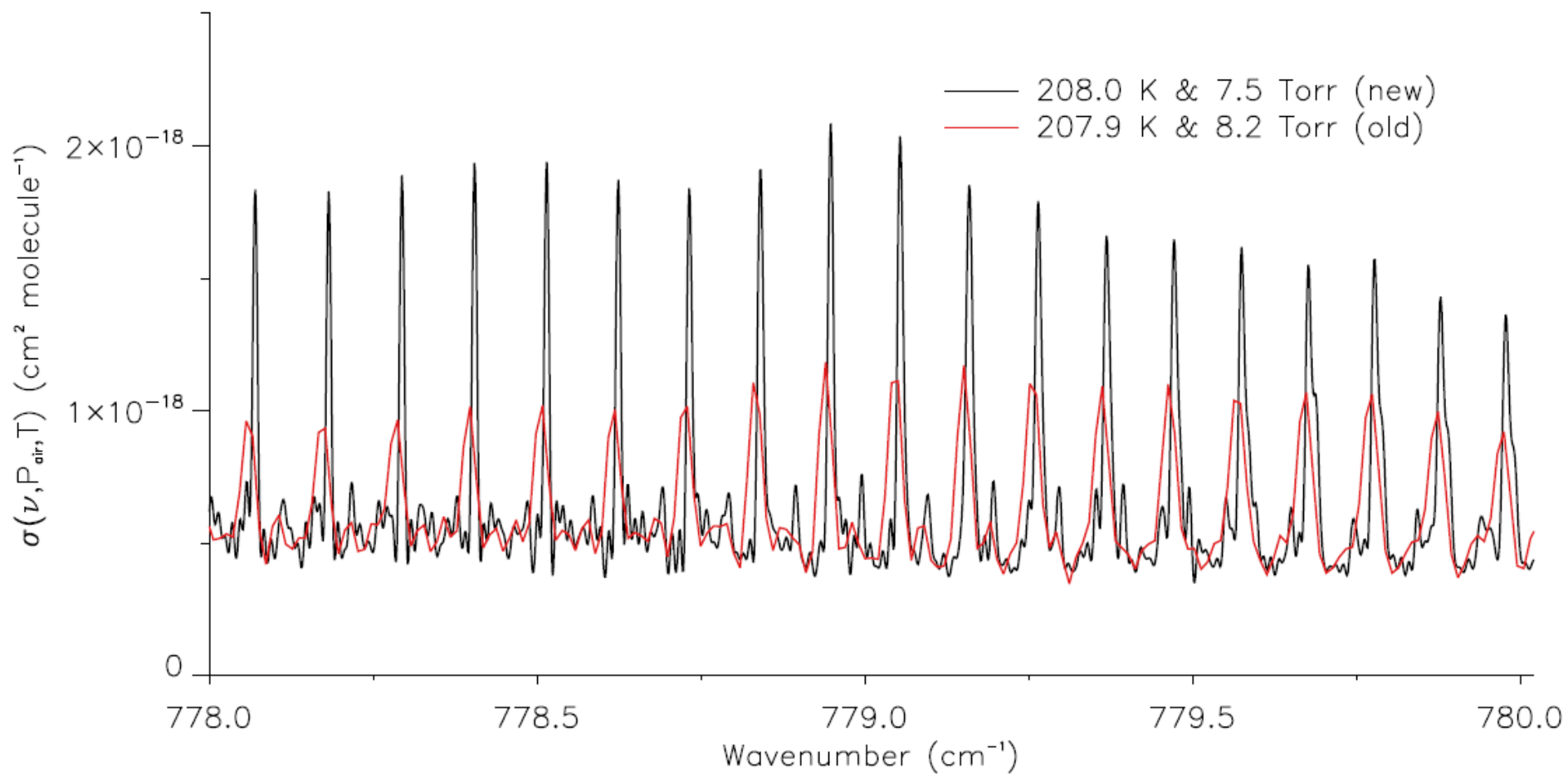


Main problems with old dataset:

- Spectra under-resolved
- Wavenumber calibration error

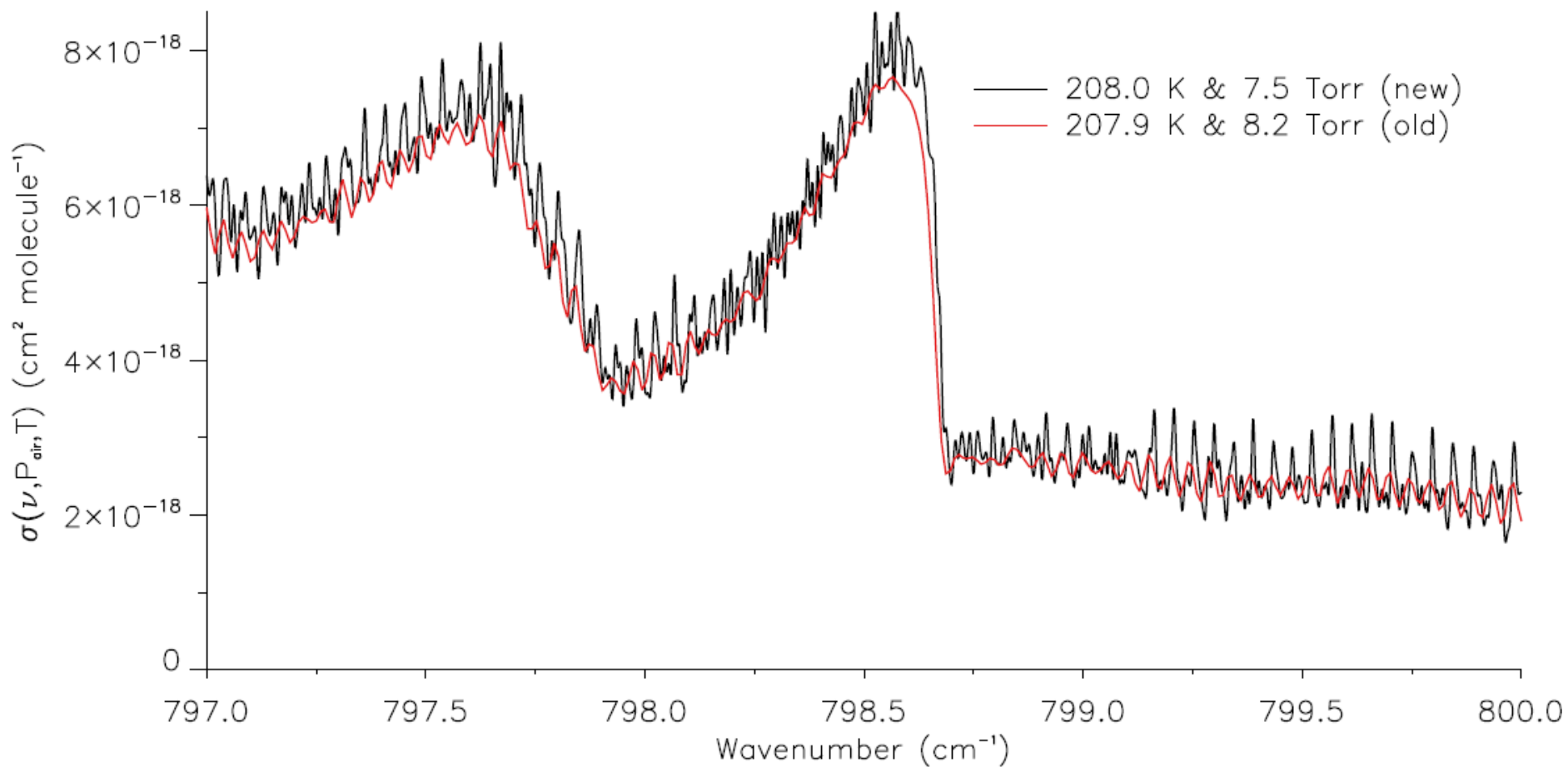


CCl_4 cross sections





CCl_4 cross sections





HCFC-22 (CHClF_2)

- Most abundant HCFC (substitutes for long-lived ozone-depleting substances)
- Lifetime 11.9 years
- HCFC-22 increased at a rate of about 8 ppt/yr (4.3%/yr) during 2007–2008, more than 50% faster than observed in 2003–2004
 - Rapidly increasing HCFC-22 production in developing countries
- Emissions are projected to begin to decline during the coming decade due to measures already agreed to under the Montreal Protocol



HCFC-22 absorption cross sections in HITRAN



Pergamon

J. Quant. Spectrosc. Radiat. Transfer Vol. 52, No. 3/4, pp. 323–332, 1994

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0022-4073(94)E0031-Z

SPECTRAL ABSORPTION-COEFFICIENT DATA ON HCFC-22 AND SF₆ FOR REMOTE-SENSING APPLICATIONS

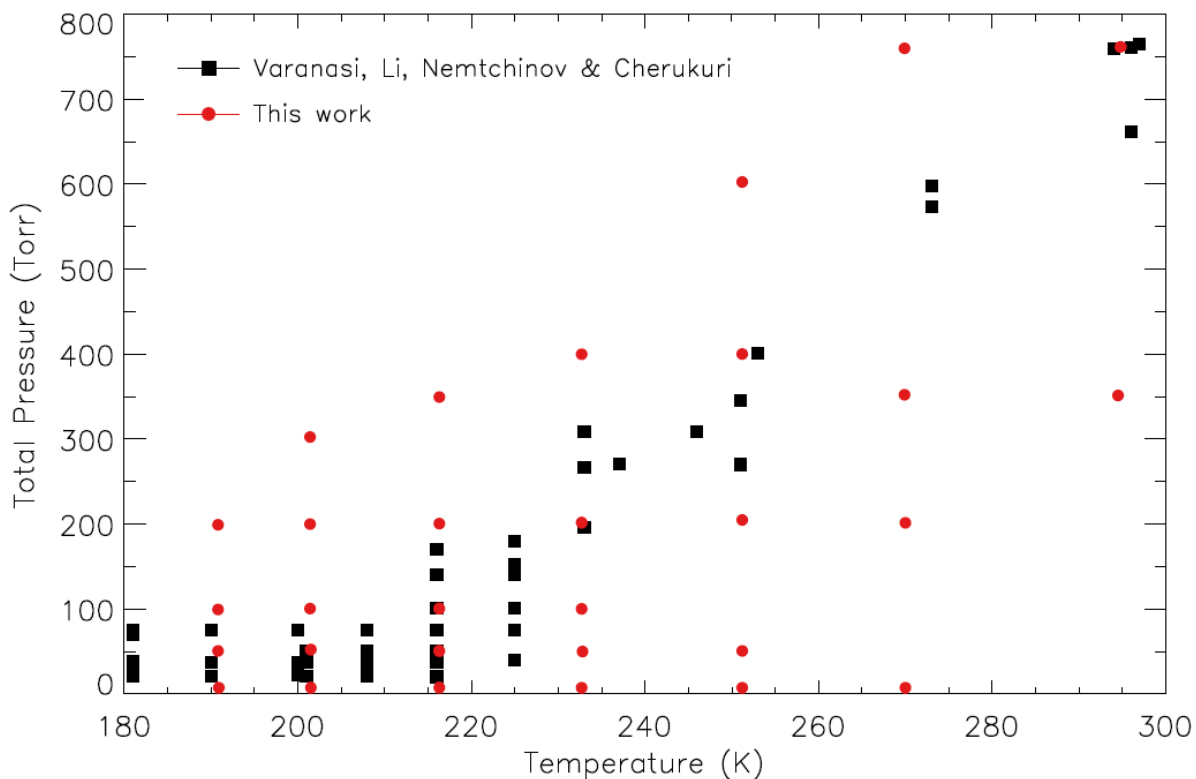
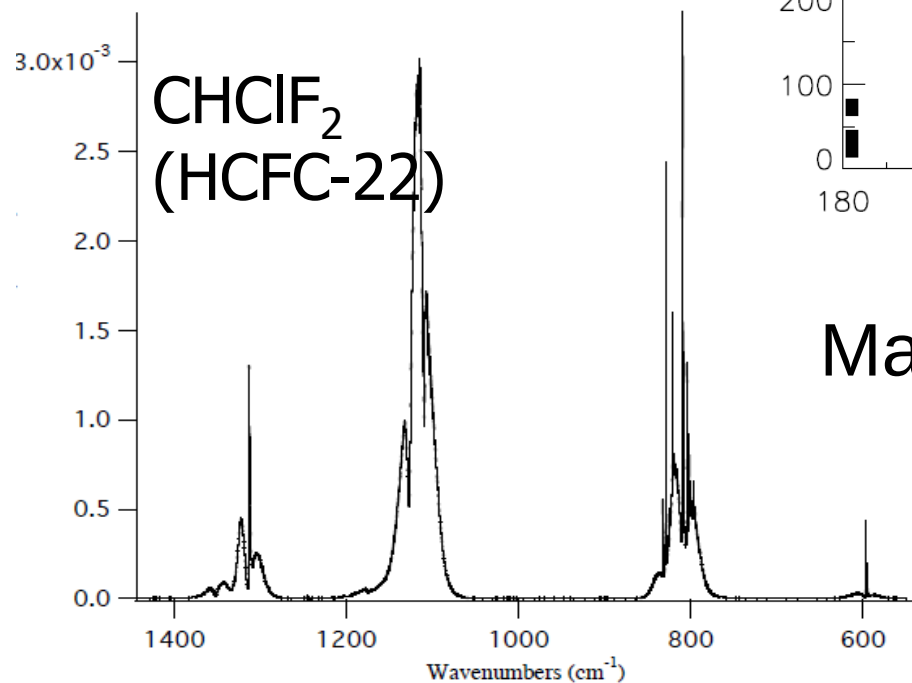
P. VARANASI,[†] Z. LI, V. NEMTCHINOV, and A. CHERUKURI

Institute for Terrestrial and Planetary Atmospheres, Marine Sciences Research Center,
State University of New York at Stony Brook, Stony Brook, NY 11794-5000, U.S.A.

Abstract—Spectral absorption-coefficients (cross-sections) k_ν ($\text{cm}^{-1}\text{atm}^{-1}$) have been measured in the 7.62, 8.97, and 12.3 μm bands of HCFC-22 (CHClF_2) and the 10.6 μm bands of SF₆ employing a high-resolution Fourier-transform spectrometer. Temperature and total pressure have been varied to simulate conditions corresponding to tropospheric and stratospheric layers in the atmosphere. The k_ν are compared with values measured by us previously using a tunable diode laser spectrometer and with the appropriate entries in HITRAN and GEISA, two of the databases known to the atmospheric scientist. The measured absolute intensities of the bands are compared with previously published values.



CHClF₂ cross sections



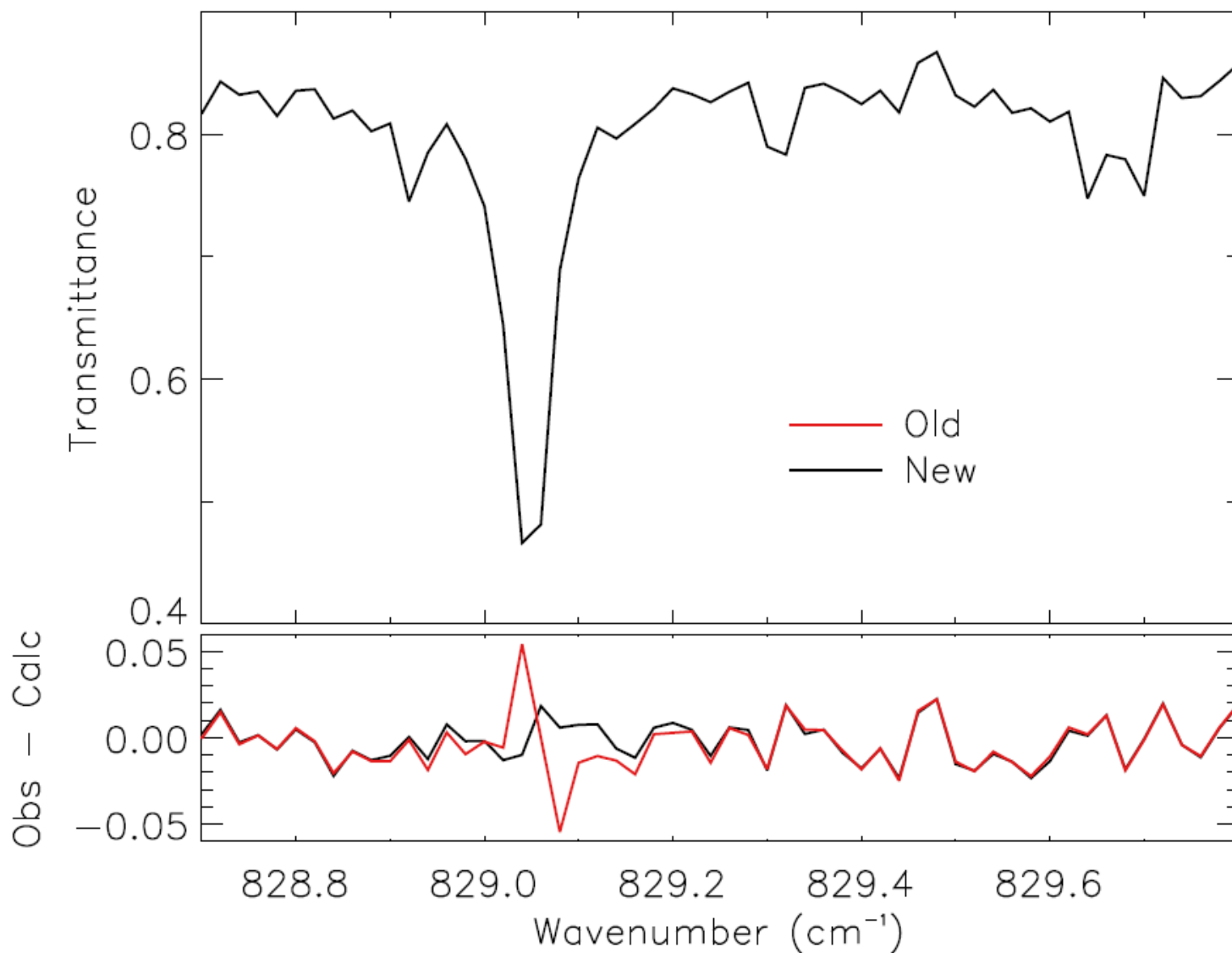
Main problems with old dataset:

- Poor P-T coverage
- Wavenumber calibration error



CHClF₂ cross sections

ACE Occultation ss11950: 7.4 km





Conclusions

- New absorption cross section datasets for HFC-23, HFC-134a, CCl_4 , CFC-12, HCFC-22
 - Will lead to better ACE retrievals
 - Will be contributed to HITRAN once published
- New measurements already taken for CF_4 , CH_2Cl_2 , CHCl_3
 - Data still being processed
- Beware when using old absorption cross section datasets!



Acknowledgements

- Funding from NERC (UK) & NCEO
- RAL – Gary Williams & Robert McPheat
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- ACE – Chris Boone & Peter Bernath